

Thank you for all the valuable comments and suggestions! Our supplementary tables and figures can be found as follows.

RESPONSE TO REVIEWER K468

Question 1: Fixing $E + H$ layers. Table 1 shows the results for LISA with only tuning $E + H$ layers. The hyperparameters are the same as the ones in the paper, and the experiments were conducted three times with different seeds.

Table 1: Results for LISA with only tuning $E + H$ layers.

MODEL	METHOD	MMLU (5-SHOT)	AGIEVAL (3-SHOT)	WinoGRANDE (5-SHOT)	MT-BENCH
TINYLLAMA	VANILLA	25.50	19.55	59.91	1.25
	FT	25.62 $\pm$ 0.10	21.28 $\pm$ 0.07	62.12 $\pm$ 0.15	2.21 $\pm$ 0.16
	LISA ($E + H$)	25.49 $\pm$ 0.14	20.75 $\pm$ 0.21	60.43 $\pm$ 0.19	2.18 $\pm$ 0.31
	LISA	26.02 $\pm$ 0.13	21.71 $\pm$ 0.09	61.48 $\pm$ 0.08	2.57 $\pm$ 0.25
MISTRAL-7B	VANILLA	60.12	26.79	79.24	4.32
	FT	61.70 $\pm$ 0.13	28.07 $\pm$ 0.12	78.85 $\pm$ 0.12	4.64 $\pm$ 0.12
	LISA ($E + H$)	61.49 $\pm$ 0.12	27.66 $\pm$ 0.07	77.93 $\pm$ 0.11	4.51 $\pm$ 0.27
	LISA	62.09 $\pm$ 0.10	29.76 $\pm$ 0.09	78.93 $\pm$ 0.08	4.85 $\pm$ 0.14
LLAMA-2-7B	VANILLA	45.87	25.69	74.11	3.29
	FT	45.66 $\pm$ 0.09	27.02 $\pm$ 0.10	75.06 $\pm$ 0.13	4.75 $\pm$ 0.16
	LISA ($E + H$)	45.88 $\pm$ 0.12	25.82 $\pm$ 0.15	73.48 $\pm$ 0.22	4.63 $\pm$ 0.35
	LISA	46.21 $\pm$ 0.12	26.06 $\pm$ 0.08	75.30 $\pm$ 0.11	4.94 $\pm$ 0.14

RESPONSE TO REVIEWER kABp

Weakness 3: LISA + LoRA experiments. For the LISA + LoRA experiment, we followed the same memory efficiency experiment in section 4.1. We set the LISA active layers to 4 and the LoRA rank to 1024, resulting in a memory cost of around 23.6 GB, which aligns with our main experiment. The learning rate for LISA + LoRA was searched within the same range described in Appendix B.3, and the experiments were conducted three times with different seeds. Figure 1 shows the training loss comparison for LISA + LoRA against the other baselines, which shows that the convergence property deteriorates in this approach. Table 2 presents the performance comparison for LISA + LoRA with other baselines for the LLaMA-2-7B model.

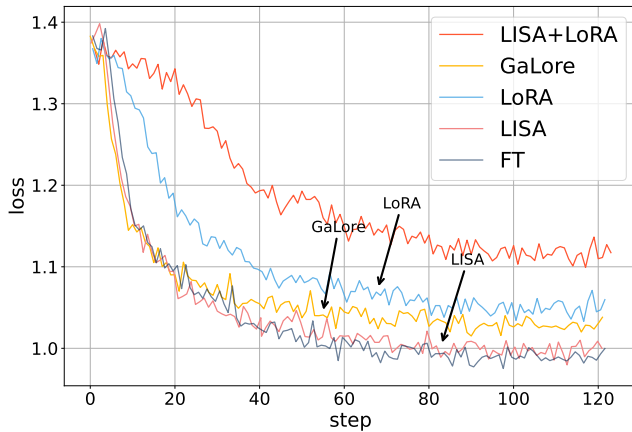


Figure 1: Loss comparison for LISA + LoRA with other baseline.

Table 2: Results for LISA + LoRA.

MODEL	METHOD	MMLU (5-SHOT)	AGIEVAL (3-SHOT)	WinoGRANDE (5-SHOT)
LLAMA-2-7B	FT	45.66 $\pm$ 0.09	27.02 $\pm$ 0.10	75.06 $\pm$ 0.13
	LoRA	45.50 $\pm$ 0.07	24.73 $\pm$ 0.04	74.74 $\pm$ 0.09
	GaLORE	45.56 $\pm$ 0.05	24.39 $\pm$ 0.11	73.32 $\pm$ 0.12
	LISA + LoRA	45.34 $\pm$ 0.41	25.55 $\pm$ 0.66	72.64 $\pm$ 0.24
	LISA	46.21 $\pm$ 0.12	26.06 $\pm$ 0.08	75.30 $\pm$ 0.11

Weakness 4: Unintuitive results. Table 3 presents the main results along with the error bars across multi-trial experiments with three different seeds.

Table 3: Multi-seed results with error bars.

MODEL	METHOD	MMLU (5-SHOT)	AGIEVAL (3-SHOT)	WinoGRANDE (5-SHOT)	MT-BENCH
TINYLLAMA	VANILLA	25.50	19.55	59.91	1.25
	FT	25.62 $\pm$ 0.10	21.28 $\pm$ 0.07	62.12 $\pm$ 0.15	2.21 $\pm$ 0.16
	LoRA	25.81 $\pm$ 0.07	19.82 $\pm$ 0.11	61.33 $\pm$ 0.09	1.90 $\pm$ 0.14
	GaLORE	25.21 $\pm$ 0.06	21.19 $\pm$ 0.07	61.09 $\pm$ 0.12	2.61 $\pm$ 0.17
	LISA	26.02 $\pm$ 0.13	21.71 $\pm$ 0.09	61.48 $\pm$ 0.08	2.57 $\pm$ 0.25
MISTRAL-7B	VANILLA	60.12	26.79	79.24	4.32
	FT	61.70 $\pm$ 0.13	28.07 $\pm$ 0.12	78.85 $\pm$ 0.12	4.64 $\pm$ 0.12
	LoRA	61.78 $\pm$ 0.09	27.56 $\pm$ 0.07	78.85 $\pm$ 0.11	4.41 $\pm$ 0.09
	GaLORE	57.87 $\pm$ 0.08	26.23 $\pm$ 0.05	75.85 $\pm$ 0.13	4.36 $\pm$ 0.16
	LISA	62.09 $\pm$ 0.10	29.76 $\pm$ 0.09	78.93 $\pm$ 0.08	4.85 $\pm$ 0.14
LLAMA-2-7B	VANILLA	45.87	25.69	74.11	3.29
	FT	45.66 $\pm$ 0.09	27.02 $\pm$ 0.10	75.06 $\pm$ 0.13	4.75 $\pm$ 0.16
	LoRA	45.50 $\pm$ 0.07	24.73 $\pm$ 0.04	74.74 $\pm$ 0.09	4.45 $\pm$ 0.15
	GaLORE	45.56 $\pm$ 0.05	24.39 $\pm$ 0.11	73.32 $\pm$ 0.12	4.63 $\pm$ 0.09
	LISA	46.21 $\pm$ 0.12	26.06 $\pm$ 0.08	75.30 $\pm$ 0.11	4.94 $\pm$ 0.14